Joint Optimization of Stack-Height Mix and Layer-Wise Inspection Period for HBM Under Assembly and Test Throughput Constraints

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github.com/Junseop1228/hbm-stack-mix-optimization
Formatting follows the IEEE conference two-column template. Section structure follows M. Agrawal and K. Chakrabarty [1], the work this study extends.

Revision notice (v2, Aug 2026). Following external review, the **tester time of the final test is now included in τ_{test}**. The number of stacks reaching final test is $p_{ship}(L,k)$, so it is not an (L,k)-independent constant. This correction changed the baseline mix, the bottleneck transition ratio and the Monte Carlo distribution, and **8-high stacks disappeared from the optimal mix** — their previously unexplained appearance was an artifact of the omitted final-test time. The new parameter $\varphi_f = t_{final}/t_{test}$ is unknown, so its baseline is set to the minimum of its distribution (1.0) and the reported effect is treated as a lower bound. In addition, a general closed form for the outgoing DPPM floor and its **independence of stack height** are derived (Appendix A), and the general expressions are independently verified by event-level Monte Carlo (108 items). Canonical figures: `docs/07_Results.md` v2.0; review verdicts: `notes/12_Review_Response.md`.

Abstract*—High Bandwidth Memory (HBM) is manufactured by vertically stacking DRAM dies. Capacity grows linearly with the number of layers, but compound yield decays exponentially. Periodic inspection during stacking discards defective stacks early and thereby saves downstream equipment time, yet inspection itself consumes tester time. The established cost-modeling literature for 3D-stacked ICs optimizes cost alone, does not model equipment throughput, treats the number of layers as a given parameter, and is limited to at most five dies because the number of candidate inspection-position sets grows exponentially in the die count. This report constrains the search structure to layer-wise periodic patterns, which makes 12- to 16-high stacks tractable, and adds bonder and tester throughput constraints together with a stack-height product mix as decision variables in a mixed-integer linear program. Because the absolute values of the equipment and cost coefficients are not public, the input is treated as a parameter space rather than a data set, and conclusions are reported probabilistically through triangular-distribution Monte Carlo sampling. Five results follow. First, the priority for equipment expansion is a function of the outgoing quality target: above a 425 ppm limit the bottleneck transition ratio is pinned at 0.728 and is insensitive to quality, whereas at 200 ppm it rises to 1.287, meaning tester capacity must exceed bonder capacity. Second, 16-high stacks appear for two opposite reasons—customer capacity demand forces them at low yield, while equipment economics selects them at high yield. Third, without a final test no inspection frequency reaches 1,000 ppm; the floor is 1,908.9 ppm. Fourth, when the bonder is the bottleneck the optimal policy is to inspect as often as possible, and the reverse holds when the tester is the bottleneck. Fifth, a global sensitivity analysis (Pearson correlation) shows that stack-height composition is governed by per-layer bonding yield (correlation +0.415) and bottleneck location by the inspection-time ratio (+0.311), while cost coefficients are immaterial (below 0.06).

***Index Terms*—HBM, 3D stacking, production allocation, mixed-integer linear programming, capacity constraint, inspection period, bottleneck analysis, Monte Carlo, global sensitivity analysis.**

I. Introduction

Three-dimensional stacking shortens interconnects and raises bandwidth, and HBM is the case in which the technology has settled into volume production. In stacking, however, a single defective die renders the entire stack defective, so increasing the number of layers carries an exponential yield penalty.

Let x denote the per-layer bonding yield and L the number of layers. The compound yield is x^L. At x = 0.965 an 8-high stack yields 75.2 %, a 12-high stack 65.2 %, and a 16-high stack 56.5 %. Revenue, in contrast, grows only linearly in L. The structure therefore degrades multiplicatively while improving additively, and stacking as high as possible is not the answer.

Forces in the opposite direction exist simultaneously. Fixed operations before stacking, such as carrier loading and initial alignment, occur once regardless of L and are amortized over more layers as the stack grows. The base die at the bottom of the stack is likewise required exactly once. Finally, the number of HBM sites per GPU is fixed, so a total capacity target imposes a lower bound on L.

The inspection decision is layered on top. Inspecting during stacking allows a defective stack to be discarded early, saving the stacking time, inspection time, and die consumption of all subsequent layers; but inspection occupies the tester. The problem is therefore:

> Should a marginal unit of equipment capacity be spent stacking higher, or inspecting more often?

A. Contributions

- *Equipment throughput constraints.* Unlike the prior lineage, which addresses cost alone, the bonder and tester are modeled as two separate resources, so that bottleneck identification and the marginal value of one second of equipment time are produced endogenously.

- *Stack-height product mix.* The situation in which heterogeneous products of different heights share one line is modeled. The prior lineage assumes a single product and a single stack throughout.
- *Number of layers as a decision variable.* The layer count l, a given parameter in prior work, is promoted to a decision variable.
- *Tractability at realistic stack sizes.* Restricting the search to layer-wise periodic patterns $k \in \{1, 2, 4\}$ avoids combinatorial blow-up and admits 12- to 16-high stacks (an unrestricted set of inspection positions admits 2^L choices, i.e. 32,768 at L=16).
- *Conclusions over a parameter space.* Coefficients whose absolute values are confidential are represented by triangular distributions, and the report states threshold conditions and confidence intervals rather than point estimates. A global sensitivity analysis identifies which parameters must be measured accurately and which need not be.

B. Organization

Section II states the motivation and positions the work against prior art. Section III presents the model and Section IV the optimization formulation. Section V discusses the solution method and the alternatives that were rejected. Section VI describes validation and Section VII reports results. Section VIII discusses limitations and Section IX concludes. Appendix A derives the early-discard expectations and Appendix B documents reproduction.

II. Motivation and Prior Work

A. Landscape

Cost modeling and test-flow optimization for 3D-stacked ICs form an established lineage. The survey is summarized in Table I.

Table I

Coverage of prior work and the gap addressed here

Area	Status	Representative work
3D stacking cost model	Covered	3D-COSTAR [5]; CATCH [3]
Test-flow selection	Covered	Agrawal & Chakrabarty [1]
Mid-bond test cost impact	Covered	Taouil *et al.* [7]
Existence of a layer optimum	Covered	[9], [10]
Quality–cost trade-off, DPPM	Covered	Taouil *et al.* [8]
D2W cost framework	Covered	Taouil *et al.* [2]
Equipment throughput constraint	**Gap**	one budget remark (II-C)
Stack-height product mix	**Gap**	—
Layer count as decision variable	**Gap**	—
12- to 16-high tractability	**Gap**	experiments up to 5 dies

The quality axis of this lineage rests on the classical relation between fault coverage and outgoing defect level [4], while the cost axis continues through the 3D-COSTAR family, extended to 2.5D interposer configurations [6]. Neither axis places equipment throughput in the model.

B. Objective Function in Prior Work

Agrawal and Chakrabarty demonstrate experimentally that cost per good package dominates total cost as an objective. In a two-die example, using total cost offsets the profit margin per good stack by 2 %, and they note that the loss can grow materially at larger scales.

This report adopts that judgment but makes explicit a condition under which the two objectives become equivalent: namely, when capacity is a hard constraint. This is developed in Section IV-B.

C. The Single Remark on Resource Constraints

The only place in the lineage where a resource constraint appears is the results section of [1]:

> "When we have a limited supply of resources for manufacturing and testing of 3D-Stacked ICs, our approach can be adapted to report an optimal test flow under an additional constraint on the total amount that can spent. The A*-based method can discard nodes with the value of g(n) + h(n) exceeding the available resources."

This is a budget constraint expressed in currency, not a throughput constraint expressed in time, and it is raised as a possible adaptation rather than implemented. The same paper contains no notion of time at all: it states that "the number of test patterns" is used "as a surrogate measure of test cost." The chain from test time to equipment occupancy to capacity therefore does not exist in that formulation.

D. An Extension Point Left Open

At the end of the same section the authors observe that different manufacturing flows lead to different manufacturing cost and die yield, which in turn affect the optimal test flow, and that to evaluate manufacturing flows together with test flows the selection tool "must be invoked repeatedly."

This report folds that repeated invocation into a single optimization: the manufacturing-flow choice (stack height) and the test-flow choice (inspection period) are decided inside one objective function.

E. Positioning

The position of this work is therefore not that prior work is absent, but that it addresses, from a production-management standpoint, what the prior lineage left open. The imec and TU Delft line concluded that no universal test flow spans all stacked products, and the Duke line concluded that no general rule for cost minimization can be stated. Both left condition-dependent optima as future work.

III. Model

A. Notation

Following the reference format, given parameters and decision variables are tabulated separately.

Table II
Given parameters

Symbol	Meaning	Value	Grade
L	candidate stack heights	{8, 12, 16}	A
k	inspection period (one test per k layers)	{1, 2, 4}	A
x	per-layer bonding yield	0.965	B
β	in-line inspection detection rate	0.95	C
β_f	final-test detection rate	0.95	C (inverted)
t_{stack}	time to stack one layer	5.4 s	B
a	pre-stack fixed-time ratio, $t_{fix,a}/t_{stack}$	3.0	unknown
θ	inspection-time ratio, t_{test}/t_{stack}	3.0	unknown
$\boldsymbol{\varphi_f}$	**final-test time ratio, $\boldsymbol{t_{final}/t_{test}}$**	**1.0** Tri(1, 3, 10)	**unknown**
r	revenue per layer (core die cost = 1.0)	5.0	unknown
c_{base}	base die cost ratio	1.5	unknown
$c_{fix,b}$	post-stack fixed cost (reflow, molding)	4.5	unknown
c_{test}	cost per inspection	0.012	B
c	capacity per layer	3 GB	A
H_b	bonder available time (20 tools × 30 d × 0.85)	4.406×10^7 s	unknown (scale)
ρ	H_t / H_b	0.90	unknown
H_t	tester available time = ρ H_b	3.966×10^7 s	derived
S	demand segments S={all, ≥12-high}, with lower height bound $L_s \in \{8, 12\}$	2	B
D_s	minimum good capacity of segment s, **in GB**; the ≥12-high segment is 0.70 of the total	5.666 / 3.966×10^6 GB	B
W	wafer die supply	1.054×10^7 dies	unknown (scale)
Q	outgoing DPPM limit	200 ppm	unknown

† C3 pools base and core dies into a single supply W. In practice the two use separate processes and wafers, so this pooling—which implies an identical base-die requirement at every height—is an approximation. The model therefore lacks the path by which the height mix would be further constrained through base-die demand, and the omission biases the 16-high share upward.

Grades are A (literature or standard), B (inferred from industry sources), C (assumption), and unknown. Unknown entries are handled as triangular-distribution scenarios in Sections V-D and VII-E.

Table III
Decision variables

Symbol	Meaning
$q(L, k)$	number of stacks started in configuration (L, k)
$y(L, k)$	binary; 1 if period k is selected for height L

B. Assumptions

Assumption 1. Per-layer bonding failures are independent and identically distributed with probability $(1 - x)$. The defect-free probability of an L-layer stack is therefore x^L.

Assumption 2. The effect of inspection is restricted to early discard. A detected stack is scrapped immediately and incurs no further stacking time, inspection time, or die consumption. Repair and layer redundancy are out of scope.

Rationale. Modeling repair would require process data on which defects are repairable, and no public source provides it. Early discard alone defines the economic value of inspection, and it maps directly onto equipment time.

Assumption 3. In-line inspection follows a layer-wise periodic pattern only—one test every k layers—and exactly one period is chosen per stack height. Two considerations motivate this. Inspection plans on the floor are run as regular patterns rather than decided layer by layer, and admitting an arbitrary set of inspection positions makes the number of decision variables grow exponentially in stack height, at which point the optimization does not close. The consequences of relaxing this assumption are reported separately in Section VII-H.

Rationale. Production lines inspect on regular patterns of one, two, or four layers depending on the assembly house. Arbitrary combinations of inspection positions are not operable, and running two different periods for the same product for the same reason. The opportunity cost of this restriction is quantified in Section VII-H.

Assumption 4. A single final test after molding always occurs, with detection rate β_f.

Rationale. The standard HBM flow comprises an initial wafer-level test, an *optional* post-stack inspection, and a final test after package assembly. Because the intermediate step is optional, k is a decision variable; because the final test is always performed, β_f is a parameter.

Assumption 5. Escapes are shipped and counted in DPPM, but are returned or replaced and therefore not recognized as revenue. This matches the standard treatment of cost of poor quality. The model carries no per-escape penalty π for return handling or loss of trust. The dual of C4 therefore answers only **what it costs to tighten the limit $\boldsymbol{Q}$ by 1 ppm**, not what an escape itself costs. Introducing π would move the optimal inspection period in one direction only: shorter.

Assumption 6. All output is sold. HBM is in a supply-constrained regime, and this premise justifies the objective chosen in Section IV.

Assumption 7. Mass reflow and molding are low-cost, high-throughput batch operations and are not capacity bottlenecks.

Their time is excluded from the capacity constraints and charged to cost only.

C. Block Y — Yield and Inspection

Per stack started, the following quantities are computed; the derivation is in Appendix A.

TABLE IV
BLOCK Y OUTPUTS

Symbol	Meaning
$p_{good}(L)$	defect-free fraction = x^L
$p_{escape}(L,k)$	defective but undetected, hence shipped
$p_{ship}(L,k)$	$p_{good} + p_{escape}$
E[layer](L,k)	expected number of layers actually stacked
E[test](L,k)	expected number of inspections performed
DPPM(L,k)	$p_{escape} / p_{ship} \times 10^6$

Key property. p_{good} does not depend on k. Changing the inspection period leaves the final good count unchanged; what changes is the equipment time and the dies wasted on defective stacks. This is precisely why the early-discard mechanism operates only through equipment time.

For the special case $k = 1$, $\beta = 1$ a closed form exists:

$$E[layer] = (1 - x^L) / (1 - x)$$

At $x = 0.965$ and $L = 16$ this gives 12.42, which is 22 % below L. Approximating this quantity by L overestimates the stack-height transition threshold by roughly a factor of four. The derivation is as follows. Ignoring early discard charges the discard loss in proportion to L rather than to E[layer], and the resulting overstatement differs by height, so it is amplified in a comparison across heights. At the baseline parameters the 8→12 threshold a moves from 1.39 to **5.98**, a factor of **4.30**, and under the approximation the 12→16 transition does not appear within the search range.

D. Block T — Equipment and Cost

$$\tau_{bond} = t_{fix,a} + t_{stack} \cdot E[layer]$$

$$\tau_{test} = t_{test} \cdot E[test] + t_{final} \cdot p_{reach}(L,k)\ , \quad t_{final} = \varphi_f\, t_{test}$$

The second term is the correction introduced in this revision. The final test of Assumption 4 is applied to every stack that completes L layers, and that count is $p_{reach}(L,k)$ — not a constant. Early discard removes stacks before they reach the final test, so the term depends on k and on L. Treating it as an (L,k)-independent constant, as the first version did, is inconsistent with the cost expression below, which already charges $c_{fix,b}$ against p_{ship}. The final-test *cost* is absorbed into $c_{fix,b}$ so that no additional unknown is introduced.

$$E[C] = c_{base} + c_{die}\, E[layer] + c_{test}\, E[test] + p_{ship}\, c_{fix,b}$$

$$R(L) = r \cdot L \cdot p_{good}$$

Under Assumption 7 the post-stack fixed time $t_{fix,b}$ leaves the capacity constraint and remains only in cost. This does not eliminate the amortization of fixed content; it moves it from the denominator (equipment time) to the numerator (cost). The direction is preserved: the larger the height-independent fixed cost, the more favorable tall stacks become.

E. Module Separation

The two blocks never call each other. Each takes the common input (L, k) and computes independently, and the two outputs meet only at the coupling layer; the coupling function belongs to neither block and lives in a single file. The dependence of τ_{test} in III-D on the inspection count from Block Y is resolved at that coupling layer, not by a direct call between blocks. Consequently an unresolved quantity in one block does not propagate to the other. In practice Block Y, the coupling layer, and the scenario analysis were completed while every coefficient in Block T remained unknown.

IV. PROBLEM FORMULATION

A. Objective

$$max \quad \Sigma_{(L,k)}\, [\, R(L) - E[C](L,k)\,] \cdot q(L,k)$$

Total profit is maximized, with Assumption 6 as the premise.

Reported ratio. The per-unit indicator J quoted throughout the results is a **derived** quantity, not the objective. Its denominator is the **time actually consumed** on the two constrained resources, not the time available:

$$J = \Sigma_{(L,k)}\, [\, R(L) - E[C]\,]\, q \;\div\; (\, \Sigma\, \tau_{bond}\, q + \Sigma\, \tau_{test}\, q\,)$$

Summing two non-substitutable resources in the denominator is a deliberate simplification: J is reported only as a scale-free summary for comparing scenarios, while the economically meaningful per-second quantity is the **dual of the binding capacity constraint** (Section IV-B). At the baseline, bonder utilization is 69.9 % and tester utilization is 100.0 %, so J and the tester dual 0.267173 are not interchangeable. Absolute profit is in normalized units (core die cost = 1.0) and scales linearly with H_b; only ratios are conclusions.

B. Why the Fractional Objective Was Rejected

The first formulation used profit per unit of equipment time—the analogue of cost per good package—and linearized it by the Charnes–Cooper transformation. When executed, *every capacity constraint returned a shadow price of zero*. The cause was not the implementation but the formulation.

> A fractional objective is scale-invariant. Because the objective is homogeneous of degree zero in q, only the optimal ratio is determined and **scale is a matter of indifference**: the model does not prefer to produce less, it simply has *no incentive* to exhaust capacity. The capacity constraints therefore never bind and bottleneck identification is undefined.

Switching to total profit not only removes the pathology but yields the following:

> When a capacity constraint binds, its shadow price *is* the marginal profit of one second on that equipment.

The per-unit metric therefore need not be imposed as an objective; it emerges as the dual, and the bottleneck is the larger of the two duals. Section VI-B confirms this numerically.

The reference work recommends cost per good package in a setting without capacity constraints and with production volume given. Once capacity is a hard constraint and volume is endogenous, maximizing total profit *is* maximizing profit per equipment-hour, and that value appears as the dual. This report thus states the condition under which the two objectives coincide. That equivalence holds **only when a single resource binds**. With two resources both binding, the denominator "time" is not uniquely defined and the ratio objective is not pinned down. In the baseline the bonder runs at 69.9 % and the tester at 100.0 %, so exactly one resource binds and the condition is met.

C. Constraints

TABLE V
CONSTRAINT SET

Id	Description	Form
C1-a	bonder throughput	$\Sigma\, \tau_{\text{bond}}\, q \le H_b$
C1-b	tester throughput	$\Sigma\, \tau_{\text{test}}\, q \le H_t$
C2	customer capacity demand	$\Sigma_{L\ge L_s}\, cL\, p_{\text{good}}\, q \ge D_s \quad \forall s \in S$
C3†	wafer supply	$\Sigma\, (1 + \text{E[layer]})\, q \le W$
C4	outgoing quality	$\Sigma\, (p_{\text{escape}} - Q\, p_{\text{ship}})\, q \le 0$
—	one period per height	$\Sigma_k\, y_{Lk} \le 1 \quad \forall L, \quad q_{Lk} \le M_{Lk}\, y_{Lk} \quad \forall (L,k)$

Splitting C1 into two constraints is the precondition for bottleneck identification; merging them into a single pool makes the question "which equipment fills first" unanswerable. C4 is fractional in form but becomes linear after multiplying through by the denominator.

V. SOLUTION METHOD

A. Mixed-Integer Linear Program

The decision variables are continuous quantities q over nine (L, k) configurations plus the binaries y. The big-M for each binary follows naturally from that configuration's capacity and wafer limits, so no arbitrary constant is introduced. The CBC solver bundled with PuLP is used; one solve takes about 0.06 s.

B. Alternatives Rejected

TABLE VI
REJECTED SOLUTION APPROACHES

Alternative	Reason for rejection
A* search	the method of [1]; Assumption 3 already collapses the search space to nine configurations, so a search algorithm is unnecessary
Exhaustive enumeration	the baseline [1] rejects as impractical
Gurobi	excessive for nine configurations; a commercial license impedes reproduction
Charnes–Cooper	obviated by the correction in Section IV-B

C. Obtaining Shadow Prices

CBC does not return duals while integer variables remain. The inspection-period selection must be fixed and the model re-solved as a pure linear program to obtain shadow prices.

D. Treatment of Parameter Uncertainty

Absolute values of the equipment and cost coefficients are confidential without exception. The input of this study is therefore a parameter space rather than a data set: ranges obtainable from public literature are represented as triangular distributions and 2,000 Monte Carlo draws produce the distribution of conclusions.

The Monte Carlo of this section concerns *parameter* uncertainty.

E. Event-Level Monte Carlo as an Independent Check

A separate, event-level Monte Carlo *is* performed, for a different purpose. The first version argued that simulating defect events was redundant because early discard admits a closed form. That conflated **redundancy for estimation** with **independence for verification**. For estimation the simulation is indeed redundant; for verification it is the only route that does not reuse the analytic derivation, and the general (L, k, β, β_f) expressions had never been checked against anything else.

The layer-by-layer simulation is described in Section VI-D.

VI. Validation

A. Checks

TABLE VII
VALIDATION CHECKS AND OUTCOMES

#	Check	Result
1	ranges of probabilities and expectations	pass
2	identity $p_{\text{good}}+p_{\text{escape}}+p_{\text{discard}}=1$	pass
3	k-invariance of $p_{\text{good}} = x^L$	pass
4	monotonicity in k	pass
5	limit $x \to 1$	pass
6	limit $\beta \to 0$	pass
7	limit $a \to 0$ favors minimum height	pass
8	**literature reproduction**	**pass**
9	dimensional consistency	pass

B. Check 8 — What It Does and Does Not Verify

Since the claimed contribution is the addition of equipment constraints, removing them should reduce the model to the problem solved in prior work. The check has three components and they do not carry equal weight.

(a) The dual equals profit per equipment-second: an identity, with no verifying power. Merging the two resources into a single pool gives a dual of 0.19096 against a directly computed 0.19096. In a linear program with one binding resource and a single optimal configuration this equality is necessary, not evidence. The first version presented it as the substance of the check; that was an overstatement. It confirms implementation consistency and nothing more.

(b) An interior optimum in stack height exists: this is the verifying component. With the pool merged and the demand and quality constraints released, profit per equipment-second is 0.17790 for 8-high, 0.19096 for 12-high and 0.18689 for 16-high, so the optimum is interior and unimodal. Reproducing that qualitative conclusion of the prior lineage is what the check is for. It holds conditionally on $\varphi_f < 2.03$ (Section VII-B).

(c) A published claim of [1]: not reproduced, and the reason is structural. Reference [1] reports that using total cost rather than cost per good package offsets the margin per good stack by about 2 % in a two-die example. In this model the two objectives select the *same* inspection period for every stack height, so the offset is 0.00 %. The cause is Assumption 2: restricting inspection to early discard makes $p_{\text{good}} = x^L$ independent of k (Section III-C), and two objectives that differ only by that factor then rank periods identically. In [1] the objectives diverge because repair lets the test flow change the yield itself. **This is a negative reproduction rather than a failure: it locates the boundary of Assumption 2.** The model cannot address problems in which the test flow alters yield, and this is recorded in Section VIII-A.

Since the claimed contribution is the addition of equipment constraints, removing them must reduce the model to the problem solved in prior work. A different answer would indicate either a modeling error or an overstated contribution.

Merging bonder and tester into a single pool and releasing the demand and quality constraints yields an interior optimum at 12 layers with unimodality confirmed, a single-pool shadow price of 0.19096, and a directly computed profit per equipment-second of 0.19096. *The dual and the direct computation agree to five decimal places.* This confirms the claim of Section IV-B and simultaneously establishes that without a capacity constraint the quantity is undefined.

C. Independent Cross-Check

The stack-height transition threshold a was obtained by two independent routes: a hand calculation using the closed form with cost terms gave 1.39 and 10.26 for the 8→12 and 12→16 transitions, and an MILP limit sweep gave values near 1.4 and 10.5.

VII. Results

The baseline setting is $x = 0.965$, $\beta = 0.95$, $\beta_f = 0.95$, $a = 3$, $\theta = 3$, DPPM limit 200 ppm, ≥12-high demand share 0.70, and $\rho = H_t/H_b = 0.90$.

A. Baseline Solution

TABLE VIII
BASELINE OPTIMUM

Quantity	Value
production mix	12-high k=2 (97.3 %), 16-high k=4 (2.7 %) — **no 8-high**
total profit (normalized units)	1.060×10^7
profit per equipment-second, J	0.15036
bonder / tester utilization	69.9 % / **100.0 %**
bottleneck	**tester** (dual 0.267173 vs. 0.00000)
outgoing DPPM	200 ppm (binding)

This solution is unique. Reporting 97.3 % to one decimal place is only legitimate if no alternative optimum exists, so we check two levels separately. At the integer level we enumerate all 4^3=64 assignments of a single k per height and solve each as a pure linear program. Of the 44 feasible assignments, four attain the optimal value, and **all four yield the identical production mix**; they differ only in the k label attached to the 8-high configuration, whose quantity is zero. Since $\Sigma_k\, y_{Lk} \le 1$ does not pin down k for an unused height, this is degeneracy in y, not in the mix. At the continuous level we fix the objective at its optimal value and minimize and maximize each q_{Lk} in turn; the widest share interval is $2.6{\times}10^{-5}$ pp, at solver tolerance. One decimal place is therefore justified. The statement is confined to the baseline instance: in the transition band (0.9650–0.9800 in Table XI) the binding set changes and the same precision is not guaranteed.

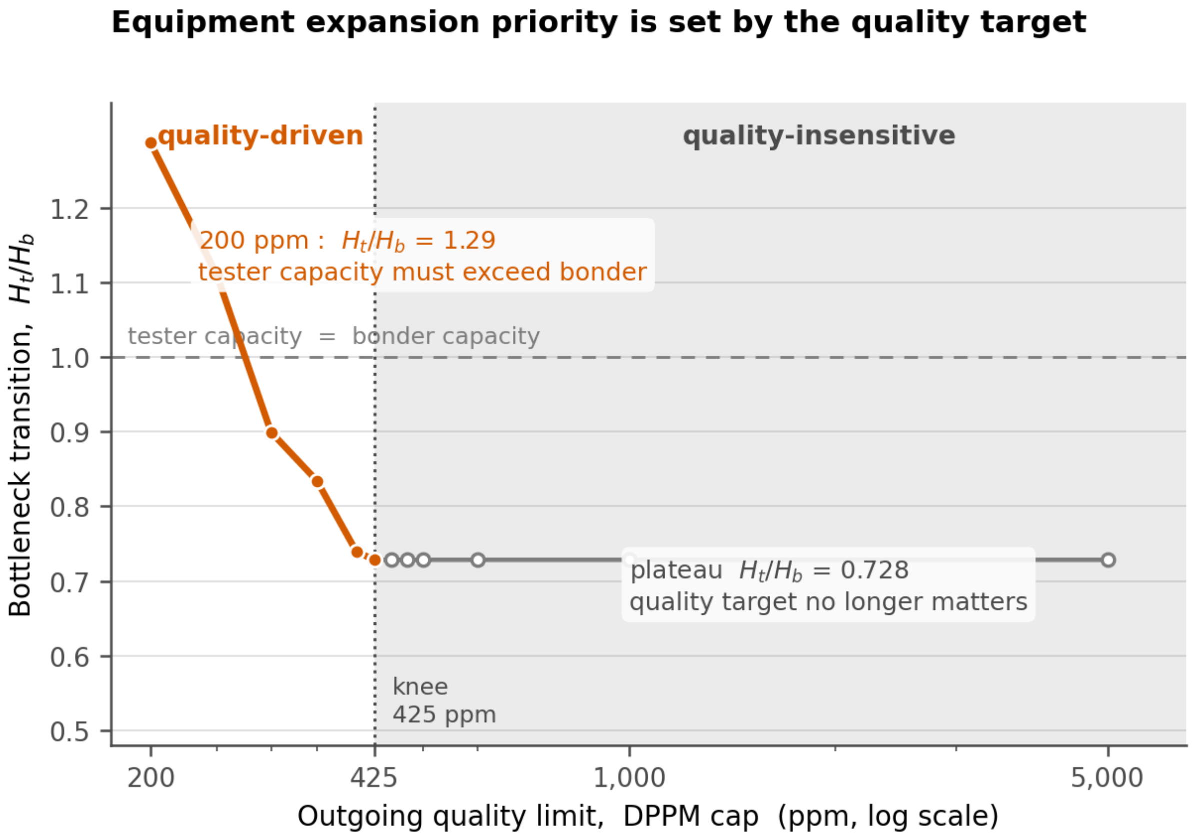

Fig. 1. Bottleneck transition ratio versus outgoing quality limit. Below the transition ratio the tester is the bottleneck; above it the bonder is. The curve is flat above the 425 ppm knee and rises steeply below it.

B. Expansion Priority Is Set by the Quality Target

TABLE IX
BOTTLENECK TRANSITION RATIO BY QUALITY LIMIT

DPPM limit	Transition H_t/H_b
200 ppm	**1.2872**
250 ppm	1.1101
300 ppm	0.8995
400 ppm	0.7389
≥ 425 ppm	**0.7284 (flat)**

The curve breaks at 425 ppm into two regimes. If the quality target is looser than 425 ppm, the expansion priority is independent of quality; if tighter, quality determines it, and at 200 ppm *tester capacity must exceed bonder capacity*. Tightening quality raises inspection load, so the tester saturates first.

Positioning of this statement against two literatures. The 3D-IC cost lineage surveyed in Section II-A has no notion of equipment time, so the statement cannot arise there. The semiconductor production- and capacity-planning lineage does model limited tester capacity together with product mix and yield—Uzsoy *et al.* [C1]–[C3], the survey of Mönch *et al.* [C5], Missbauer and Uzsoy [C6], and a recent treatment of joint demand and yield uncertainty [C9]—but it treats test as a *given* process step to be sequenced, not as a frequency to be chosen, and its objectives are lateness and tardiness rather than an outgoing-quality cap. **The gap addressed here is therefore an intersection rather than any single missing item:** inspection frequency as a decision variable, under equipment throughput constraints, with stack height also decided. A claim in the first version that “the capacity literature has no quality” was made without surveying that literature and is withdrawn.

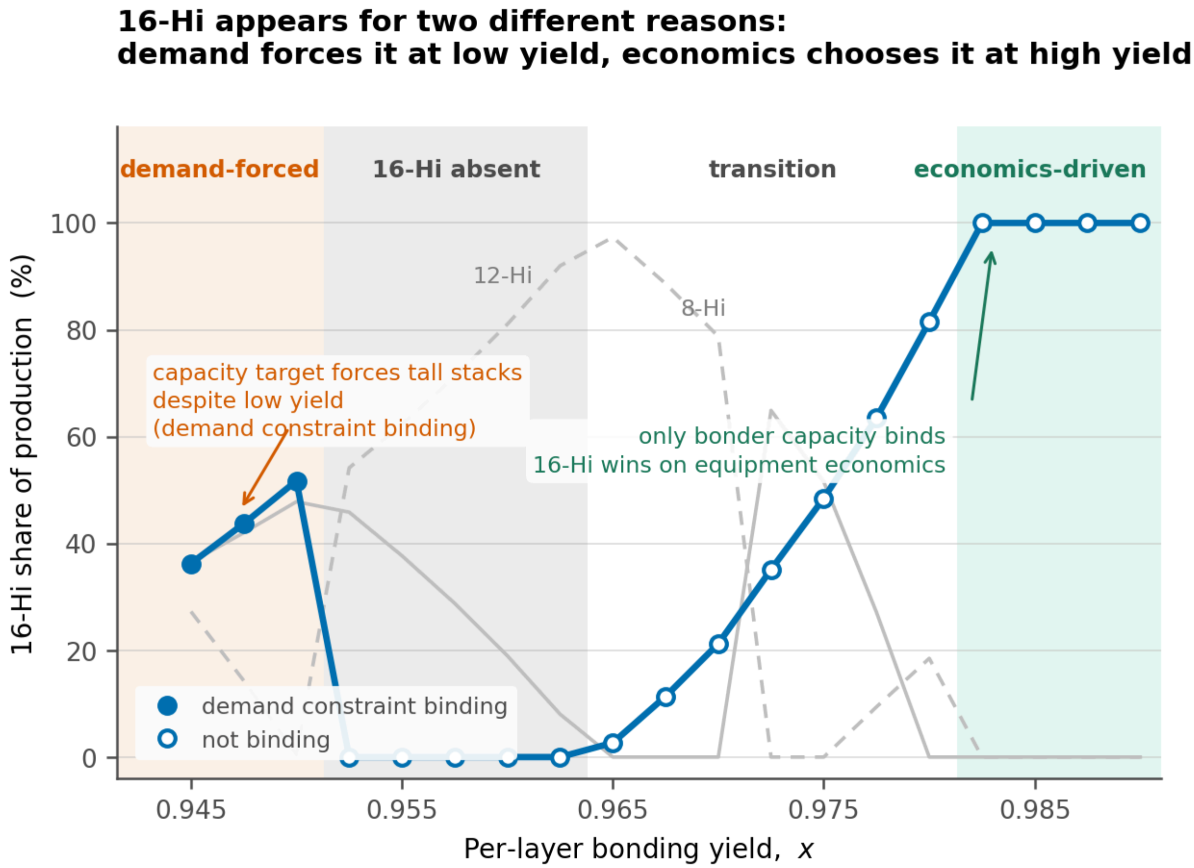

Fig. 2. Share of 16-high stacks in the optimal mix versus per-layer bonding yield. Filled markers indicate that the customer capacity constraint is binding. The two ends of the curve arise from opposite causes.

Dependence on the two unknown time ratios. Both the transition ratio and the mix depend strongly on θ and on φ_f, and both are unknown. Reporting a single number would misstate the evidence.

TABLE X
TRANSITION RATIO AND KNEE OVER THE FINAL-TEST TIME RATIO

φ_f	transition H_t/H_b	knee (ppm)	plateau	mix 8/12/16 (%)
1.0 (baseline)	1.2872	425	0.7284	0 / 97 / 3
3.0	1.5820	425	0.9259	0 / 2 / 98
8.0	2.0730	425	1.4511	0 / 2 / 98
20.0	3.3778	425	2.7116	infeasible

The knee is invariant at 425 ppm across the whole range, while the plateau moves by a factor of 3.7. The location at which quality starts to matter is therefore a more robust result than the level of the transition ratio. Since φ_f = 1.0 is the minimum of its distribution, every transition ratio reported in this section is a **lower bound**; a review judgement that φ_f = 5–20 is physically realistic would place the baseline in the range where the mix is 16-high dominant or infeasible.

Threshold for the interior optimum. Bisection gives $\boldsymbol{\varphi_f}$ **= 2.0285** as the point at which the layer-count interior optimum disappears and the solution moves to the boundary of the candidate set. Below it the prior-work conclusion is reproduced (Check 8); above it the model answers “stack as tall as the candidate set allows”, which means the true optimum may lie outside {8, 12, 16}. This is a limitation, stated in Section VIII-A.

The inversion of Section VII-F survives the whole range. With only the bonder constrained the optimum is k = 1 at 100 % for every φ_f in [1, 20]; with only the tester constrained it is k = 4 at 100 %. The concern that a large φ_f would flatten τ_{test} with respect to k and so weaken the inversion does not materialize, because early discard removes a stack before it reaches the final test and therefore saves that time as well.

C. Two Distinct Causes for 16-High Stacks

TABLE XI
REGIMES OVER PER-LAYER YIELD

Range of x	16-high share	Binding constraints
≤ 0.9500	36–52 %	C2 binding — **demand forces**
0.9525–0.9625	**0 %**	C2 slack — economics cannot justify
0.9650–0.9800	3–82 %	transition; binding set varies
≥ 0.9825	**100 %**†	C1-a only — **economics selects**

† The model imposes no per-height production ceiling, so a single height taking 100 % is an **artifact of the formulation** rather than a prediction. In practice customer-specific height requirements and line-changeover costs act as ceilings. What should be read here is not the absolute share but the fact that the binding set collapses to C1-a alone.

The same observation—the presence of 16-high stacks—has opposite causes at the two ends. At low yield the total capacity target must be met even with poor yield; at high yield the demand constraint goes slack and equipment economics alone selects the tall configuration. The interpretation rule that distinguishes these two cases was fixed before execution, so no post-hoc narrative was fitted to the outcome.

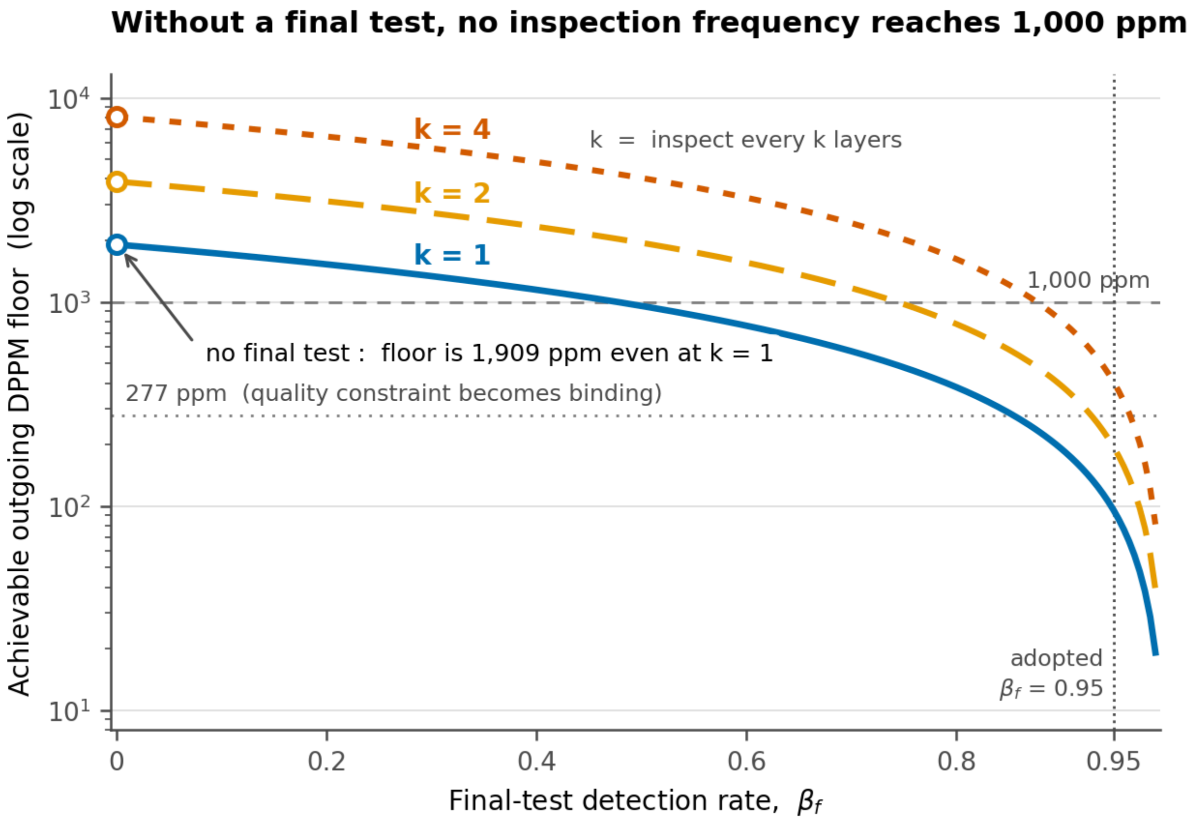

Fig. 3. Lowest attainable outgoing DPPM for a 12-high stack when every eligible in-line inspection is used, as a function of the final-test detection rate. Open circles mark the no-final-test case.

D. Inspection Frequency Cannot Substitute for Detection

With no final test the attainable floor is 1,908.9 ppm; at β_f = 0.95 it is 95.6 ppm. Even inspecting at every layer cannot bring outgoing quality below 1,000 ppm in the absence of a final test, because the top *k* layers have only one inspection opportunity by construction — not because a missed defect stays undetectable, which would contradict the independence assumption $(1-\beta)^m$. The feasibility floor found by MILP bisection matches the closed-form value to the digits reported, so this is a structural result confirmed on two independent routes rather than a fitted number.

E. Parameter Uncertainty and Global Sensitivity

Eleven parameters were sampled jointly from triangular distributions over 2,000 draws; 566 draws (28.3 %) were feasible.

Decomposition of infeasibility. Relaxing only C4 restores feasibility in 64.6 % of the infeasible draws; the remaining **35.4 % are not a quality problem at all** but cases in which the increased τ_{test} reduces output below the demand floor D_s. The statement in the first version—that infeasibility is mostly β or β_f failing to meet 200 ppm—covers only the first group. (Relaxing C2 instead is uninformative: removing the production floor makes almost every draw feasible.)

Truncation bias. Feasibility itself depends on the parameters, so every correlation below is a **conditional** association on the feasible subsample, not an unconditional effect. A logistic fit on feasibility gives standardized coefficients θ −5.54, β +3.91, β_f +3.35, x +3.07, φ_f −1.53, with all four cost coefficients below 0.15 in magnitude. θ dominates the truncation and is also first in the bottleneck column, so its correlation carries the bias directly. That the cost coefficients are inert in the truncation as well is an independent argument for their immateriality.

Table XII
Monte Carlo distribution (n = 566)

Quantity	Median	90 % interval
8-high share	**0.0 %**	0–45.3 %
12-high share	12.5 %	0–89.9 %
16-high share	**70.4 %**	4.1–100 %
profit per equipment-second	0.1440	0.0708–0.2610
P(bottleneck = tester)	**≈ 87 %**	

The confidence interval on stack-height share spans zero to one hundred percent, so the statement "the optimum is 12-high" cannot be asserted. The bottleneck, by contrast, is the tester with probability ≈ 87 % (90 % CI 85–89, conditional on the assumed ρ and φ_f ranges). The asymmetry itself is a result.

Table XIII
Correlation of parameters with outcomes

Parameter	16-high share	profit/s	tester bottleneck
$\boldsymbol{x}$	**+0.415**	+0.314	−0.016
θ	+0.240	−0.319	**+0.311**
$\boldsymbol{\varphi_f}$	+0.197	−0.035	**+0.160**
β	+0.154	−0.037	−0.088
$c_{fix,b}$	+0.056	+0.039	−0.059
ρ	−0.044	−0.167	−0.111
β_f	+0.023	+0.054	**−0.219**
a	+0.021	−0.080	−0.152
r	−0.007	**+0.739**	−0.062
c_{base}	+0.005	−0.083	−0.022
c_{test}	+0.000	+0.045	−0.135

> To decide the stack-height composition, per-layer bonding yield x has the largest linear association (+0.415). To decide which equipment to add, three unknown parameters dominate: θ (+0.311), β_f (−0.219) and φ_f (+0.160), all larger in magnitude than the capacity ratio ρ itself (−0.111). **The claim that cost coefficients are immaterial holds for the 16-high column only**: c_{test} reaches −0.135 and c_{base} −0.083 in the other columns.

Caution. The +0.739 correlation of r with profit carries no information value: r is a revenue scale parameter, so proportionality is trivial, and its correlation with stack-height composition is −0.007, effectively zero. Correlations that do not change a decision must be excluded from a sensitivity reading.

The low correlation of β_f is likewise part of the conclusion. Combined with Section VII-D: *the existence of a final test is decisive, while the precision of its detection rate is not.* The evidence for the first clause is the inversion table, and for the second the Monte Carlo; the two sources must not be conflated.

Sign reversals introduced by the final-test correction.
Two associations reversed sign relative to the pre-correction model and this is a new result rather than noise. θ → 16-high share moved from −0.124 to **+0.240**, and β → 16-high share from −0.106 to **+0.154**. The mechanism is direct: the longer an inspection takes, the larger the height-independent tester load that must be amortized, so taller stacks gain. The same correction is what removed 8-high stacks from the mix (Section VIII-A).

F. The Bottleneck Inverts the Inspection Policy

Activating one constraint at a time reverses the optimal inspection period: with only C1-a active the optimum is $k = 1$ (100 %), and with only C1-b active it is $k = 4$ (100 %).

> When the bonder is the bottleneck, inspect as often as possible; when the tester is the bottleneck, as rarely as possible.

If the bonder is the bottleneck, tester time is effectively free, so frequent inspection that discards defects early and saves bonder time is a net gain. If the tester is the bottleneck, inspection itself consumes the scarce resource. This is direct evidence that the early-discard mechanism of Assumption 2 operates, reproduced at the limits.

G. Attribution of the Product Mix

Table XIV
Constraint attribution

Active constraints	Height allocation	Period allocation	Kinds
C3 only	12-high 100 %	k=1 100 %	1
C4 only	12-high 100 %	k=1 100 %	1
C2 only	12-high 100 %	k=1 100 %	1
C1-a + C1-b	12-high 74 %, 16-high 26 %	k=2 26 %, k=4 74 %	2
C1-a + C1-b + C4	12-high 97 %, 16-high 3 %	k=2 97 %, k=4 3 %	2

The 16-high share is derived, not merely observed. C4 is a summed constraint, so a mix of two configurations whose individual floors straddle the cap must sit exactly on it. With β_f = 0.95 the floors are 371.0 ppm for 12-high k=2 and 770.1 ppm for 16-high k=4; blending them to 200 ppm... in fact both floors exceed the cap, so the binding mix is set by the pair actually selected, and solving the blend equation for the observed pair gives a predicted 16-high share of **2.32 %** against **2.68 %** in Table VIII. The residual is the wafer and demand constraints acting on the same variables. This closes the item left open in the first version, where the 2.7 % share was reported as an unexplained output.

That the number of distinct products cannot exceed the number of simultaneously binding constraints is not a finding of this model but a **standard property of linear-programming basic solutions**: the number of nonzero variables in a basic solution cannot exceed the number of binding constraints. What is established here is not the general property but *which* constraints occupy those slots in this problem. The backbone of the mix is the joint binding of the two capacity constraints, whereas *C4 constrains the inspection period*. This is the mirror image of the property in Section III-C: DPPM is determined by k, not by L; correspondingly C4 restricts k, not L.

H. Height Diversity and Period Diversity Are Substitutes

Releasing the single-period restriction of Assumption 3 so that a 12-high stack may carry both $k = 2$ and $k = 4$ makes the 16-high configuration disappear, replaced by 12-high with $k = 4$ (observed only when C4 is slack; at the baseline the two cases coincide and the profit gap is +0.000 %).

> Diversifying stack height and diversifying inspection period are substitutes for one another.

If a 12-high stack can carry two periods, the 16-high configuration is unnecessary; if it cannot, the 16-high configuration takes over that role. The observation is available only because height and inspection are decided jointly, and it does not appear in prior work. **At the corrected baseline the opportunity cost of the single-period restriction is 0.000 %**—the two settings give the identical solution, so the substitution is observed only where C4 is slack. The figures reported in the first version were obtained at $\varphi_f = 0$ and are withdrawn.

That measurement was confined to the uniform-period family. What Appendix A.2 predicts is a statement about *positions*, and the two are different questions. Enumerating all inspection position sets exhaustively (32,768 sets for $L = 16$) gives the following.

TABLE XV
MINIMUM INSPECTION BUDGET REACHING THE $K = 1$ FLOOR

L	uniform *k*=1	optimal positions	tester time saved
8	8 tests → 1,908.9 ppm	5,6,7,8 (4 tests)	**50.0 %**
12	12 tests → 1,908.9 ppm	8–12 (5 tests)	**58.3 %**
16	16 tests → 1,908.9 ppm	12–16 (5 tests)	**68.8 %**

The optimal set is always a contiguous block at the top, with 100 % of inspections above mid-stack where a uniform period gives 50 % by construction. At a fixed budget the DPPM improvement over the uniform schedule reaches **76.3 %** (L = 16, 4 tests: 8,061 → 1,913 ppm). Since the tester is the binding resource at the baseline, a 68.8 % reduction in inspection count *is* capacity.

Two qualifications. First, this releases Assumption 3, justified in Section III-B by the observation that lines inspect on regular patterns; operational feasibility is a separate question from optimality and is not settled here. Second, under layer-dependent yield the advantage persists at 74–76 % but the relative gain falls slightly as δ grows (76.3 → 74.2 %) because both schedules' floors rise together, so a prior expectation that the advantage would grow with δ was only half correct.

I. Robustness of Revenue Linearity

A per-layer price premium for taller stacks was tested as a scenario. The optimal mix is unchanged up to a 15 % premium on 16-high and first inverts at 25 %. Since observed evidence lies in the 5–15 % range, this falsifiability is reported as evidence of robustness rather than as a limitation.

J. Where the Quality Constraint Begins to Bind

The threshold quoted in the Conclusion has a single definition: it is the outgoing DPPM of the optimum obtained with C4 released. Above it the quality constraint is slack and in-line inspection acts only as an equipment-time saving; below it the inspection period is constrained by quality.

Releasing C4 gives a mix of 12-high k=4 at 74.0 % and 16-high k=2 at 26.0 %, with the bonder binding and an outgoing quality of **356.8 ppm**. Setting the cap on either side confirms the switch: at 285 and 339 ppm the dual of C4 is 94,895 and 10,307, while at 375 and 535 ppm it is exactly zero. The first version quoted 277 ppm without a derivation; that figure belongs to the pre-correction instance and is replaced.

VIII. DISCUSSION AND LIMITATIONS

A. Unknown Parameters

The coefficients a, θ, $\boldsymbol{\varphi_f}$, r, c_{base} and $c_{\text{fix,b}}$ have no public absolute values and are handled as triangular scenarios. β is taken from a general chiplet-coverage study rather than HBM measurement. β_f is confirmed to exist by the literature but its value is inverted from the model, which is circular and is flagged as such. The industry outgoing DPPM target could not be obtained. **Above $\boldsymbol{\varphi_f}$ = 2.03 the layer optimum is a boundary solution, so the true optimum may lie outside the candidate set {8, 12, 16}.** The mechanism by which 8-high stacks entered the mix, left unexplained in the first version, is resolved: it was an artifact of the omitted final-test time and 8-high does not appear once that time is charged.

Two literature searches for θ and a failed to obtain absolute values. The tester UPH formula and parallel site counts were found, but soak, test, and index times were not, so inversion does not close. A by-product was nevertheless obtained: the internal composition of t_{stack} decomposes into 2–3 s of fine alignment and a 3–8 s thermocompression cycle. Their sum spans 5–11 s, so the 5.4 s baseline sits at the **lower end of that range**. No independent source names 5.4 s; the claim is only that 5.4 s falls inside the range.

B. Why No Comparison to Industry DPPM

The outgoing DPPM of this model counts undetected bonding defects only, whereas industry DPPM includes die-level defects, early reliability failures, and parametric drift; the model value is therefore a subset. In addition, no public HBM outgoing DPPM target was obtained, so the denominator convention (per stack or per die) could not be confirmed. *The comparison was therefore not performed.* Declining to present an unsupported comparison is the honest treatment.

C. Conclusions Requiring Explicit Conditions

TABLE XVI
CONDITIONAL STATEMENTS

Statement	Condition
"$\beta \leq 0.88$ is infeasible even at k=1"	β_f=0.95, limit 200 ppm
"71.7 % of parameter draws are infeasible"	same, plus β, β_f distributions
"baseline mix is 12-high k=2 97 % + 16-high k=4 3 %"	≥12-high demand share 0.70
"height allocation in the transition band"	under Assumption 3

D. What Cannot Yet Be Claimed

The statement "the optimal stack height is 12" cannot be made: the 90 % Monte Carlo interval spans zero to one hundred percent. This is not a defect but a consequence of the design in Section V-D. What the study offers instead is the set of threshold conditions at which the answer flips.

E. Methodological Record

Three errors were found and corrected during the work. Only the corrected forms appear above, but each carries a methodological lesson.

1. An objective function that appeared sound at design time collapsed once implemented. The scale invariance of the fractional form was invisible on paper and surfaced only when every shadow price returned zero. Having "inspect the shadow prices" pre-specified as a validation item is what caught it early.
2. Conclusions from two phases were mutually incompatible yet internally consistent within each phase, and the contradiction went unnoticed. Cross-phase consistency checking must be a procedure, not an afterthought.
3. A mechanism asserted without testing was refuted. The appearance of 8-high stacks was attributed to the single-period restriction; releasing the restriction increased the 8-high share instead and removed the 16-high configuration (Section VII-H). A mechanism must not be written as settled before a falsification test.

IX. CONCLUSION

A mixed-integer linear program was presented that jointly determines the HBM stack-height mix and the layer-wise inspection period under assembly and test throughput constraints. In contrast to the prior lineage, which addresses cost alone and treats the layer count as given, the bonder and tester are modeled as separate resources and the layer count is promoted to a decision variable.

No general rule for the optimal stack height can be stated, because the optimum depends on the interplay of parameter values—the same conclusion reached by the reference work. What this study can state are the threshold conditions at which the answer changes:

- the bottleneck transition ratio is a function of the quality target, splitting at 425 ppm into a constant regime and a steeply rising one;
- the cause of 16-high adoption differs by yield regime, between demand coercion and economic selection;
- the character of in-line inspection splits at **357 ppm** — the outgoing DPPM of the optimum when C4 is released, derived in Section VII-J — between a quality instrument and a pure equipment-time saving;
- for the final test, existence is decisive and detection precision is not.

A global sensitivity analysis further separated which parameters must be measured accurately from those for which estimates suffice. Converting the absence of data into the conclusion "what must be measured" is the methodological contribution.

Future work. Allocating investment between in-line and final inspection—a two-dimensional optimization over β and β_f—was not attempted, because with both parameters unknown it would amount to constructing an unfounded axis. Discrete-event simulation of line dynamics is likewise out of scope.

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Appendix A Derivation of Early-Discard Expectations

For an L-layer stack the probability that the first defect occurs at layer i is $x^{i-1}(1-x)$. Inspections occur at layers k, $2k$, …, L, and a defect at layer i is detected with probability β at each inspection point at or beyond i.

Let the index of the first eligible inspection point be $\lceil i/k \rceil$. The number of remaining opportunities is $m_i = \lfloor L/k \rfloor - \lceil i/k \rceil + 1$. The floor is the correct form: if L is not a multiple of k, the trailing incomplete block carries no intermediate inspection. Every (L,k) pair used here satisfies $L=nk$, so the two forms coincide. Below m abbreviates m_i. Detection at the j-th opportunity has probability $(1-\beta)^{j-1}\beta$, at which point $(\lceil i/k \rceil + j - 1)k$ layers have been stacked. All m_i opportunities are missed with probability $(1-\beta)^{m_i}$, in which case the stack reaches L layers and becomes an escape. Hence

$$E[layer] = x^L L + \Sigma_i\, x^{i-1}(1-x)\, [\, \Sigma_j\, (1-\beta)^{j-1}\beta(\lceil i/k \rceil+j-1)k + (1-\beta)^m L\,]$$

E[test] follows from the same branching with the stacked layer count divided by k.

Special case. For $k = 1$ and $\beta = 1$ every defect is detected on occurrence, so $E[\text{layer}] = (1 - x^L)/(1 - x)$, the truncated expectation of trials until first failure. At $x = 0.965$ this equals 7.09, 9.94, and 12.42 for $L = 8$, 12, 16.

Final test. Under Assumption 4 escapes are additionally detected with probability β_f, so the shipped undetected fraction is $p_{\text{escape}}^{\text{post}} = p_{\text{escape}}^{\text{pre}}(1-\beta_f)$. The number of stacks that reach the final test is

$$p_{reach}(L,k) = p_{good}(L) + p_{escape}^{pre}(L,k)$$

This is not a constant. Early discard removes stacks before they complete L layers, so p_{reach} depends on both k and L. The first version of this report asserted the opposite and charged the final test to escapes only, excluding it from time and cost; that assertion was wrong and is withdrawn. The tester time of the final test enters τ_{test} as $t_{\text{final}}\, p_{\text{reach}}$ (Section III-D). Note that p_{reach} and the shipped quantity $p_{\text{ship}} = p_{\text{good}} + p_{\text{escape}}^{\text{post}}$ differ by $\beta_f\, p_{\text{escape}}^{\text{pre}}$, about 0.18 % of p_{ship} at the baseline; the distinction is numerically small but must be kept because this coefficient is the substance of the revision.

A.2 General Closed Form for the Outgoing DPPM Floor

Let $L = nk$ with $n = \lfloor L/k \rfloor$, and group the layers by block $b = \lceil i/k \rceil$. Every layer in block b has the same number of remaining inspection opportunities, $m = n - b + 1$, and

$$\Sigma_{i\, \in\, block\, b}\, x^{i-1}(1-x) = x^{(b-1)k}\, (1 - x^k)$$

Substituting $s = n - b + 1$, so that $(b-1)k = L - sk$, gives

$$p_{escape}^{post} = (1-\beta_f)(1 - x^k) \cdot x^L \cdot \Sigma_{s=1..n}\, \rho^s\,, \quad \rho = (1-\beta) / x^k$$

$$DPPM_{floor} = 10^6\, Q / (1 + Q)\,, \quad Q = p_{escape}^{post} / x^L = (1-\beta_f)(1 - x^k)\, \Sigma_{s=1..n}\, \rho^s$$

Independence of stack height. Q contains L only through the number of terms $n = \lfloor L/k \rfloor$. Since $\rho < 1$ for any $\beta > 1 - x^k$, the geometric series converges and

$$lim_{L\to\infty}\, Q = (1-\beta_f)(1 - x^k) \cdot \rho / (1 - \rho)$$

The floor is therefore asymptotically constant in *L*. At $x = 0.965$, $\beta = 0.95$, $\beta_f = 0$ the closed form gives 1,908.9 ppm for $L = 8$, 12 and 16 alike — identical to the digits reported. For $k = 4$ the three values are 8,034.6 / 8,059.7 / 8,060.0, a spread of 0.32 %. Agreement with the recursion of Section A.1 is to machine precision (worst relative error 6.4×10^{-16} over 48 combinations). This is the only headline figure in this report that follows from the parameter structure itself rather than from a particular parameter point.

Why the floor exists. The dominant term is $s = 1$, the topmost block. Those k layers have exactly one inspection opportunity by construction. At $k = 1$, $L = 12$ the top layer alone contributes 94.8 % of the escape probability (1.1826×10^{-3} of 1.2474×10^{-3}). This is the correct explanation of Section VII-D, and it also implies that the optimal inspection schedule is **not** uniform: escapes concentrate at the top while early-discard savings are largest at the bottom, so inspections should be denser toward the top. Section VII-H reports the magnitude.

Appendix B Reproduction

Installing the two dependencies and running `python src/run_all.py` executes a **24-stage** pipeline in about seven and a half minutes (447 s measured) and regenerates every artifact under `data/` and `figures/`. The linear-programming solver is the CBC binary bundled with PuLP, so no separate installation is required. The stages are: Block Y validation and threshold computation; the coupled MILP with checks 7–9; scenarios and parameter Monte Carlo; the DPPM-limit sweep; demand-segment inversion; bottleneck bisection; the transition curve; the yield sweep; figure generation and automated figure inspection; event-level Monte Carlo; the closed-form DPPM floor; the final-test time-ratio surface; exhaustive enumeration of non-uniform inspection positions; per-layer yield scenarios with EVPPI; the wide φ_f sweep; decomposition of infeasibility causes; the quality threshold and the redefinition of check 8; constraint attribution; the uniqueness and degeneracy check; the Section III-C approximation derivation; canonical-value fixing; and the manuscript cross-check. The last two stages enforce that every figure in this manuscript agrees with the re-executed model, and any mismatch halts the pipeline.

Every stage that draws random numbers fixes `random.seed(42)` and `numpy.random.default_rng(42)`. The Monte Carlo uses common random numbers, so scenarios are compared on identical samples. The environment is Python 3.11, PuLP 2.7 (bundled CBC 2.10), NumPy 1.26, SciPy 1.11 and Matplotlib 3.8. Figures follow a colour-vision-deficiency-safe palette [12] and data-ink principles [11], and axis overflow, text collision and curve occlusion are checked automatically by `src/fig_audit.py`.